

Zhuhan Bao

Zhejiang, China | zhuhan.bao@duke.edu | github.com/BaoZhuhan

Education

Hangzhou City University, B.Eng. in Software Engineering
GPA: 3.77/4.0 Average: 86.29/100

Sep 2023 – Jul 2027

Publications

SIMAX: A Scalable and Interpretable Framework for Multi-fidelity and Annotated Clinical Dialogue Simulation

Zhuhan Bao[†], Rui Yang[†], Bohao Yang, et al., Chuan Hong^{*}

Submitted to npj DM, in editorial review

 [arXiv:2606.30491](https://arxiv.org/abs/2606.30491)  [AutoEvaluation/SIMAX](#)

As co-first author, I developed SIMAX, a framework for generating large-scale, controlled, and multimodal clinician–patient dialogue data with reference behavioral annotations. I implemented the end-to-end pipeline integrating predefined clinical scenarios, persona and voice conditions, codebook-derived communication behavior targets, LLM-based dialogue generation, and two-speaker audio synthesis. I led the system implementation, experimental design, data generation, intrinsic data quality assessment, downstream utility evaluation, and manuscript preparation.

Toward Realistic Evaluation of Clinical Diagnostic Reasoning through Multimodal Evidence Synthesis

Rui Yang, Weihao Xuan, **Zhuhan Bao**, Huitao Li, et al., Nan Liu^{*}, Yifan Peng^{*}

Working Paper (not publicly indexed)

Developed the ClinMM-Bench to evaluate multi-turn multimodal clinical diagnosis, where I executed the reasoning evaluation experiments through atomic fact decomposition and comparison.

OS-Omni: Benchmarking Generalist Computer-Using Agents on Long-Horizon Workflows Across Platforms

Hui Shen, et al.

Working Paper, Scheduled for release in June

 [Hugging Face Dataset](#)

Designed and contributed benchmark tasks for Linux and MacOS, and participated in the review of benchmark tasks across Windows, Linux, and Android.

Research Experience

Visiting Student, Advisor: Prof. **Chuan Hong**

July 2025 – Present

Department of Biostatistics & Bioinformatics, Duke University School of Medicine
Duke University, Durham, NC, USA

Conduct research on clinical dialogue simulation, medical AI agents, and synthetic healthcare data generation. Lead the development of LLM-based simulation frameworks for behaviorally grounded doctor–patient interaction modeling, multimodal clinical communication analysis, and large-scale synthetic dataset generation.

In parallel, support the laboratory’s computational infrastructure by maintaining unified research environments on the DCC computing cluster and conducting H200 GPU benchmarking and workload evaluation for large scale AI research deployment.

Visiting Student, Advisor: Prof. **Peiyu Liu**

May 2025 – Aug 2025

Zhejiang University NESA Research Lab
Zhejiang University, Hangzhou Zhejiang China

Systematic Security Impacts Investigation of MCP Servers in AI Code Editors

Conducted research on MCP server ecosystems and their security implications in AI-assisted code editors.

Explored SDLC-oriented plugin analysis using LLMs and web crawling techniques, and developed

 MCP-Collector for automated MCP Marketplace data collection and classification.

Undergraduate, Advisor: Prof. Lin Sun
The School of Computer and Computing Science
Hangzhou City University, Hangzhou Zhejiang China

Mar 2025 – Mar 2026

Medical Multimodal Dataset Construction and Large Model Research Based on Online Teaching Videos

This project aims to construct a high-quality bilingual (Chinese-English) pathology image-text dataset from online educational videos and fine-tune the CLIP model using the Lora method to enhance the performance and accuracy of intelligent pathology diagnosis and medical knowledge retrieval. As project leader, I am mainly responsible for developing web crawlers, aligning the dataset, and fine-tuning the model.

Teaching Experience

Teaching Assistant, Advisor: Prof. Kui Su
High Performance Computing

26 Spring

Designed and graded assignments; led problem-solving sessions; assisted in homework evaluation. Topics included MPI, OpenMP, and parallel algorithms.

Project Experience

Embodied World Model Inference Optimization

Jan 2026 – Mar 2026

2026 ASC Student Supercomputer Challenge

This project focused on accelerating the inference of UnifoLM-WMA-0, a state-of-the-art embodied world model, while maintaining a PSNR threshold of ≥ 25 dB. I conducted systematic performance profiling using Nsight Systems to identify bottlenecks in GEMM kernels and memory I/O. I implemented a multi-level optimization strategy including mixed precision (FP16/TF32), Flash Attention (SDPA) integration, and CUDA Graph execution to reduce CPU launch overhead. Additionally, I developed fused CUDA kernels for DDIM updates and optimized the data pipeline using pinned memory and asynchronous H2D transfers. The final pipeline achieved a $2.68\times$ speedup over the FP32 baseline, successfully passing all 20 robotic evaluation scenarios.

RNA 5-Methylcytosine Construction Optimization

Jan 2025 – Feb 2025

2025 ASC Student Supercomputer Challenge

This project focuses on optimizing the detection workflow for RNA 5-methylcytosine (m5C) modification sites, aiming to enhance both precision and computational efficiency. I was responsible for designing and implementing automated scripts for data cleaning, alignment, deduplication, and statistical filtering, as well as tuning parameters to achieve high accuracy and reduced runtime. The project achieved significant improvements in automation, runtime, and memory usage, and established a robust, reproducible pipeline for m5C site detection.

Learning Experience

Competitions

1st National Service Computing Innovation Competition

Excellent Award

2025 ASC Student Supercomputer Challenge

Second Prize

2026 ASC Student Supercomputer Challenge

Second Prize

16th Lanqiao Cup C/C++ Programming, Zhejiang Province

Second Prize

Captain of School High-Performance Computing Lab

Mar 2025 – Mar 2026

Hangzhou City University Supercomputing Center

Captain of the Voluntary Maintenance Team

Sep 2024 – Sep 2025

Voluntary Maintenance Team of the CS School

Skills

Certificates: CET-4, CET-6, PRC Driver's License (C1)

Programming Languages: C/C++, MPI, Python, Bash, Triton, CUDA

Skills: Inference Optimization, Parallel Computing, Machine Learning, Data Analysis, Algorithm Design

Software & Platforms: Linux, Slurm, Vtuner, Perf, SSH, Docker, Git, Conda, PyTorch